

The Möbius-Pair Sequence Is Not One Multiplicative Function

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Abstract

The literal sequence $c(n)=\mu(n)\mu(n+2)$ is not multiplicative: $c(3)=c(5)=1$ but $c(15)=-1$. One-function pretentious theorems therefore cannot be applied to the product merely by renaming it. The classification is `METHOD_OBSTRUCTION_L1` and the verdict is `STOP_SCOPED`. No program-positive L2 claim or endpoint exponent credit is made.

1 Target contract and source boundary

The target has six simultaneous axes: actual fixed- h_0 packet, source-locked named physical atom, every deterministic prefix, every deterministic scale, a fixed- X power at that atom, and actual active support. The value $h_0 = 2$ is source-backed data only. Repository hashes are integrity locks, not theorem evidence. TPC-193's declared seven-source corpus remains `STOP-SCOPED` [1].

2 A determinant-two specialization

Take $D(n) = n$ and $V(n) = n + 2$, so the coefficient determinant is $1 \cdot 2 - 1 \cdot 0 = 2$. Define

$$c(n) = \mu(n)\mu(n+2).$$

Theorem 1 (Multiplicativity firewall). *The function c is not multiplicative. In particular,*

$$c(3) = 1, \quad c(5) = 1, \quad c(15) = -1,$$

although $(3, 5) = 1$.

Proof. Using $\mu(3) = \mu(5) = \mu(7) = \mu(17) = -1$ and $\mu(15) = 1$, the three values are immediate. Thus $c(15) \neq c(3)c(5)$. $\square$

3 What is and is not stopped

Pretentious distance and Halász-type theorems whose hypothesis is a single multiplicative function cannot be invoked on c after a purely notational declaration that c is multiplicative. The calculation says nothing against genuine two-function Elliott/Chowla inputs, bilinear decompositions, or theorem-backed structure special to the affine pair.

4 Loss, level and scope ledger

The smallest missing literal theorem is

`THEOREM_FOR_COUPLED_VALUES_ON_TWO_AFFINE_FORMS.`

The next route is `TWO_FUNCTION_CORRELATION_ROUTE`. The cell `ONE_FUNCTION_PRETENTIOUSNESS_DIRECT_APPLICATION_TO_CZ` is `STOP-SCOPED`; neither O161 parent nor the global architecture is stopped.

The named-atom exponent credit is 0, while the endpoint requires a strict budget greater than $1/400$; the ledger is unpaid. Block sums are not cumulative prefixes, symbolic packet atoms are not named production atoms, phase L^2 and almost-everywhere statements are not pointwise evaluation, and scoped method failures are not theorem or architecture failures.

5 Machine certificate

The adjacent canonical payload freezes the formula or finite witness, exact repository-source hashes, any explicitly non-hashed external locator record, six target axes, scoped stop, route state and claim firewall. Its recursive exact schema has closed objects, exact array positions and constant leaves. The checker recomputes the finite certificate and executes ten adversarial mutations.

References

- [1] Liang Wang. Literal fixed-atom candidate mechanism gate, 2026.